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**ARiS – AI driven Accessibility Risk Mapping and Prediction Intelligence
System (Major Project)**

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Abstract

Urban accessibility remains a critical challenge in modern cities in India, particularly for elderly individuals and individuals with mobility limitations. Traditional methods of assessing accessibility rely heavily on manual audits and static compliance checks, which are often time-consuming, limited in scope, and unable to capture complex real-world conditions. This project addresses these limitations by developing an AI-Driven Urban Accessibility Risk Mapping and Prediction Intelligence System (ARiS).

The project's main goal is to create a system that can support data-driven decision-making and assess accessibility at the road segment level. The proposed solution integrates machine learning, rule-based evaluation, and interactive visualisation into a unified framework.

Accessibility risk is predicted using a Random Forest model based on important characteristics such as sidewalk width, slope, curb ramp, traffic, lighting, and surface quality. Simultaneously, a rule-based compliance engine ensures interpretability by assessing compliance with accessibility criteria. Additional features include a confidence and uncertainty layer, an AI explanation (XAI) module, a scenario simulation module, accessibility risk prediction for new areas, a priority intervention engine for ranking segment based on urgency, and a city planning dashboard with a budget optimizer for intervention planning.

The system demonstrates consistent and reliable performance across different scenarios. High-risk segments are accurately identified based on poor infrastructure conditions, while improvements in input parameters result in measurable reductions in risk and increases in compliance. Confidence scores improve transparency by demonstrating the accuracy of predictions.

Overall, the proposed system provides a comprehensive and practical tool for urban accessibility improvement. It combines predictive capability with interpretability, enabling planners to identify critical areas, evaluate potential interventions, and prioritise resources effectively. The system has strong potential for real-world application and can be further extended through integration with live urban data sources.

Table of Contents

Abstract.....	2
1. Chapter 1: Introduction.....	6
1.1. Background and Motivation.....	6
1.2. Aims and Objectives.....	7
1.3. Scope and Contributions.....	7
2. Chapter 2: Technical Background and Literature Review.....	8
2.1. Technical Background.....	8
2.2. Industry Solutions.....	9
2.3. Literature Review.....	9
3. Chapter 3: Proposed System: Design and Implementation.....	11
3.1. System Overview.....	11
3.2. System Components.....	12
3.3. Implementation Details.....	14
4. Chapter 4: Performance Evaluation Setup and Scenarios.....	18
4.1. Evaluation Methodology.....	18
4.2. Test Environment.....	19
4.3. Test Scenarios.....	20
5. Chapter 5: Results and Discussions.....	21
5.1. Experimental Results.....	21
5.2. Performance Analysis.....	25
5.3. Discussion	28
6. Conclusions and Future Works.....	30
6.1. Summary of Contributions.....	30
6.2. Limitations	30
6.3. Future Work.....	31
References.....	32
Appendix.....	33

List of Figures

Figure 1: Use Case UML Diagram.....	11
Figure 2: Activity UML Diagram.....	12
Figure 3: Data Preparation code snippet 1.....	14
Figure 4: Data Preparation code snippet 2.....	14
Figure 5: Data Preparation code snippet 3.....	14
Figure 6: Model Integration code snippet 1.....	14
Figure 7: Model Integration code snippet 2.....	14
Figure 8: Risk prediction code snippet	15
Figure 9: Compliance evaluation code snippet	15
Figure 10: Scenario Simulation code snippet 1.....	15
Figure 11: Scenario Simulation code snippet 2.....	15
Figure 12: Model Comparison code snippet	16
Figure 13: Confidence Layer code snippet 1.....	16
Figure 14: Confidence Layer code snippet 2.....	16
Figure 15: Priority Ranking code snippet 1.....	16
Figure 16: Priority Ranking code snippet 2.....	16
Figure 17: Accessibility Risk Prediction code snippet 1.....	17
Figure 18: Accessibility Risk Prediction code snippet 2.....	17
Figure 19: Model Insights code snippet.....	17
Figure 20: Explanation for ML Predictions.....	17
Figure 21: City Planning code snippet 1.....	18
Figure 22: City Planning code snippet 2.....	18
Figure 23: ARiS.....	20
Figure 24: Confidence score snippet 1.....	25
Figure 25: Confidence score snippet 2.....	25
Figure 26: AI risk explanation.....	26
Figure 27: Compliance snippet 1.....	26
Figure 28: Compliance snippet 2.....	26
Figure 29: Priority Engine snippet 1.....	27
Figure 30: Priority Engine snippet 2.....	27
Figure 31: Priority Engine snippet 3.....	27
Figure 32: Scenario Simulator snippet 1.....	27
Figure 33: Scenario Simulator snippet 2.....	27
Figure 34: Accessibility Risk Prediction for New Area snippet 1	28
Figure 35: Accessibility Risk Prediction for New Area snippet 2.....	28
Figure 36: Budget Optimizer.....	28

List of Tables

Table 1: TC001.....	21
Table 2: TC002.....	22
Table 3: TC003.....	22
Table 4: TC004.....	22
Table 5: TC005.....	23
Table 6: TC006.....	23
Table 7: TC007.....	23
Table 8: TC008.....	24
Table 9: TC009.....	24
Table 10: TC010.....	25
Table 11: TC011.....	25

Chapter 1: Introduction

1.1 Background and Motivation

To make cities accessible and usable for everyone, urban accessibility is crucial. People of all ages and physical capabilities, including the elderly and those with impairments, can walk freely and safely in public areas thanks to accessible infrastructure. An urban environment's accessibility is directly influenced by factors including sidewalk width, surface quality, curb ramps, slope conditions, lighting, and traffic levels.

Urban planning has begun to rely increasingly on technology-driven solutions due to the city's rapid growth and rising population density. To analyse infrastructure and facilitate smarter decision-making in city management, tools like Geographic Information Systems (GIS), data analytics, and machine learning are being used increasingly. These technologies enable the processing of massive amounts of spatial data and the discovery of patterns that would be challenging to find through manual inspection alone.

Despite the presence of accessibility guidelines and universal design standards, evaluating urban accessibility remains a challenging task in India. In many cases, accessibility evaluations are frequently conducted using physical audits or manual surveys. Although, these approaches can provide valuable insights, they are frequently time-consuming and challenging to scale throughout big cities. As a result, accessibility issues are frequently identified only after complaints are raised or problems become visible.

Another drawback of many existing methods is that they don't predict possible accessibility issues instead, they mostly concentrate on mapping infrastructure conditions. While mapping techniques can identify places with poor infrastructure, they typically don't offer planners predictive insights that help them predict potential accessibility issues.

This gap highlights the potential role of machine learning and data-driven analysis in urban accessibility studies. Machine learning models can analyse infrastructure characteristics and identify patterns associated with accessibility challenges. When combined with tools for spatial visualization, these systems can assist identify regions that might need attention before issues worsen.

The problem addressed in this project is the absence of a practical system that integrates infrastructure analysis, predictive modelling, and spatial visualization to identify accessibility risks at a street or block level. Planners, researchers, and policymakers could better understand accessibility circumstances with the help of a system that can predict and visualize such dangers.

To address this issue, this project proposes ARiS (AI-Driven Urban Accessibility Risk Mapping and Prediction Intelligence System). Using machine learning techniques, the system analyses infrastructure features and predicts accessibility risk levels. An interactive map interface is then used to illustrate these predictions, making it simple for users to investigate accessibility problems in various locations.

The motivation behind this work is to explore how data-driven technologies can contribute to more informed and inclusive urban planning. By combining rule-based accessibility checks with machine learning predictions, the project aims to demonstrate how technology can support better understanding of accessibility challenges in urban environments.

1.2 Aim and Objectives

Aim

The aim of this project is to develop a system that uses machine learning techniques to assess urban infrastructure data and predicts accessibility risk levels at the street or block level, while presenting the results through an interactive map-based visualization.

Objectives

To achieve this aim, the following objectives have been defined:

1. Identify key infrastructure features that influence urban accessibility, such as sidewalk width, slope, curb ramps, lighting conditions, surface quality, and traffic levels.
2. Create a structured dataset representing street segments or blocks in India with accessibility-related attributes.
3. Develop a rule-based scoring system based on official accessibility guidelines to evaluate infrastructure conditions.
4. Implement a supervised machine learning model capable of predicting accessibility risk levels.
5. Design and develop a web-based application that visualizes predicted risk levels using an interactive map.
6. Evaluate how effectively the system can identify and represent accessibility risks.

1.3 Scope and Contributions

Scope of the Project

The goal of this project is to develop a prototype system that uses a structured dataset representing urban infrastructure conditions in Indian cities to illustrate the idea of accessibility risk prediction. The system analyses selected infrastructure attributes and classifies street segments into different accessibility risk levels.

The scope of the project includes:

- Analysis of infrastructure features related to accessibility.
- Prediction of accessibility risk using machine learning techniques.
- Integration of rule-based accessibility evaluation.
- Visualization of risk levels through an interactive map interface.

The project does not seek to establish a real-time infrastructure monitoring system or a full-scale city-wide deployment. Instead, the focus is on creating a functional prototype that shows how a data-driven approach to accessibility risk assessment is feasible.

Contributions of the Project

The main contributions of this project include:

1. the development of a hybrid accessibility assessment approach that combines rule-based accessibility standards with machine learning-based risk prediction.
2. The creation of an interactive map-based system that shows accessibility risks at a street or block level.
3. The demonstration of how machine learning and data analytics methods can support better understanding of accessibility conditions in urban environments.

Overall, this project highlights the potential of integrating data analysis, machine learning, and map visualization to support more informed and inclusive urban planning decisions.

Chapter 2: Technical Background and Literature Review

2.1 Technical Background

Urban accessibility has become a critical focus within smart city research and inclusive urban planning. Rapid urbanisation, particularly in developing countries, has increased the need for pedestrian-friendly and barrier-free infrastructure. Traditional accessibility assessment methods often rely on physical observations and design-based evaluations, which can be time-consuming, subjective, and difficult to scale across large urban areas. Studies on walkability highlight that urban design qualities such as safety, comfort, and infrastructure significantly influence pedestrian accessibility (Ewing & Handy, 2009).

This issue is particularly significant in India, where rapid urbanisation has increased pressure on infrastructure, and national programmes such as Smart Cities Mission have emphasised the need for efficient, sustainable, and inclusive urban development (Government of India, 2024).

To address these limitations, Geographic Information Systems (GIS) have been widely adopted for map analysis and mapping of urban accessibility. GIS enables the identification of infrastructure related issues and supporting decision-making process. In the context of accessibility, such issues may include narrow sidewalks, poor connectivity, lack of curb ramps, unsafe crossings, inadequate lighting, and uneven surface conditions. Since smart city research highlights the need for integrated, data-driven, and predictive approaches to successfully manage urban challenges, modern urban systems require more than just visualization (Yigitcanlar et al., 2019).

To improve decision-making, machine learning techniques have recently been used in urban analytics. Supervised learning models, particularly ensemble methods such as Random Forest, have been shown to perform well in classification problems involving urban data due to their ability to handle multiple features and nonlinear relationships. These models can analyse various parameters such as sidewalk width, slope, curb ramp, surface conditions, lighting conditions, and traffic levels to classify accessibility risk levels across urban segments (Breiman, 2001).

When deploying AI systems, model interpretability has emerged as a crucial issue in addition to prediction. Black-box models often lack transparency, which can reduce trust among stakeholders. To address this, Explainable Artificial Intelligence (XAI) techniques have been developed. SHAP (SHapley Additive Explanations) is a popular technique that offers a reliable way to describe how each feature contributes to the model's prediction. This enhances the usability of AI systems in real-world decision-making scenarios (Lundberg & Lee, 2017).

Furthermore, multi-criteria decision-making (MCDM) approaches have been applied in urban planning to prioritise interventions. These methods combine multiple factors such as infrastructure condition, safety risks, and social importance to rank areas based on urgency and impact. Such frameworks support planners in allocating resources efficiently and addressing critical areas first (Saaty, 2008).

Overall, the integration of GIS, machine learning, explainability techniques, and multi-decision-making provides a strong foundation for developing intelligent and scalable urban accessibility assessment systems.

2.2 Industry Solutions

Several industry solutions address urban accessibility and infrastructure analysis, but most focus on specific aspects rather than providing a complete, integrated system.

Geographic Information System (GIS) platforms, such as ArcGIS, are widely used for map analysis. These tools allow users to map and visualize infrastructure data, including sidewalks, and road networks. While GIS platforms are powerful for visualization, they often lack built-in predictive capabilities for accessibility risk (ESRI, 2023).

Mapping services like Google Maps provide basic accessibility information, including wheelchair-friendly routes in certain regions. However, these features are often limited in coverage, rely heavily on user-contributed data, and do not perform detailed risk analysis at a segment level (Google, 2023).

StreetLight Data, which focuses on traffic and mobility patterns, is another example. While it provides valuable insights into transportation behaviour, it does not specifically address accessibility compliance or pedestrian infrastructure quality (StreetLight Data, 2023).

Similarly, many smart city platforms integrate sensor data and analytics to monitor urban conditions. However, these systems frequently prioritize environmental monitoring or traffic control over accessibility (Batty et al., 2012).

A common limitation across these solutions is the lack of integration between prediction, compliance evaluation, and decision support. Most tools either visualise data or provide analytics, but do not combine multiple components into a unified system.

The proposed system addresses this gap by integrating machine learning prediction, rule-based compliance, explainability (XAI), a confidence and uncertainty layer, scenario simulation, budget optimizer, and priority ranking into a single platform.

2.3 Literature review

Recent research has explored the use of data-driven approaches for urban accessibility and infrastructure analysis.

Several studies have used machine learning techniques to predict urban conditions based on environmental and spatial data. These methods show that models are capable of successfully finding patterns in complex datasets. However, many of these studies focus on high-level analysis, such as city-wide trends, rather than detailed segment-level evaluation (Zheng, 2015).

Other research has focused on accessibility compliance using rule-based methods. These studies highlight the importance of standards and guidelines in evaluating infrastructure. While

these approaches are reliable and interpretable, they lack adaptability and may not capture complex interactions between factors (Imrie & Hall, 2001).

Some studies have attempted to combine machine learning with rule-based systems to balance Accuracy and interpretability. However, these hybrid approaches do not often include additional features such as confidence estimation or scenario simulation (Doshi Velez & Kim, 2017).

In recent years, explainable AI has also drawn more attention. According to research, providing explanations for model predictions improves trust and reliability, especially in decision-support systems. Techniques such as feature importance and model interpretation are commonly used to achieve this (Lundberg & Lee, 2017).

A limitation observed in existing research is the lack of end-to-end systems. Most studies focus on individual components, such as prediction or analysis, without integrating them into a complete workflow (Yigitcanlar et al., 2019).

The proposed system builds upon these ideas by combining multiple approaches into a single framework. It not only predicts accessibility risk but also evaluates compliance, provides explainability (XAI), incorporates a confidence and uncertainty layer, simulates infrastructure improvements, and prioritises interventions to support decision-making.

Comparison of Existing Solutions

The table below summarises key differences between existing approaches and the proposed system:

Approach / System	Prediction Capability	Compliance Evaluation	Explainability	Priority ranking	Scenario Simulation
GIS Platforms (e.g., ArcGIS)	No	limited	Low	No	No
Mapping Apps (Google Maps)	No	Basic	Low	No	No
Traffic Analytics (e.g., StreetLight Data)	Partial	No	Low	No	No
Rule-Based Systems	No	Yes	High	Limited	No
ML-Based Research Models	Yes	No	Low/Medium	No	No
Proposed System (ARiS)	Yes	Yes	High	Yes	Yes

Summary

This chapter reviewed the key technologies and research relevant to the project. It highlighted the strengths and limitations of existing approaches and identified gaps in current solutions.

The analysis shows that while individual components such as machine learning, rule-based evaluation, and GIS tools are well-developed, there is a lack of integrated systems that combine these elements effectively.

The proposed system addresses this gap by providing a comprehensive solution that combines prediction, evaluation, explainability, and decision support. This forms the basis for the system design presented in the next chapter.

Chapter 3: Proposed system – Design and Implementation

3.1 System Overview

The ARiS (Accessibility Risk Mapping and Prediction Intelligence System) is a proposed system designed to evaluate accessibility problems at the road-segment level. This system focuses on identifying risks at a deeper level, allowing for more accurate and practical urban planning decisions, in contrast to standard methods that assess accessibility at a larger city scale.

Use Case Diagram

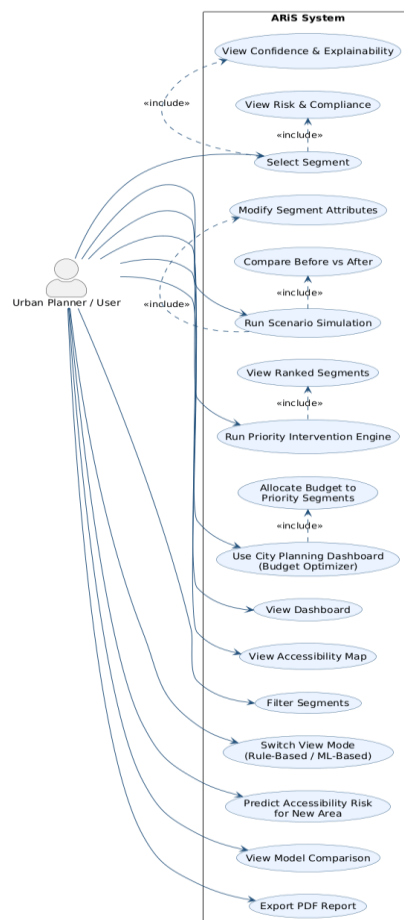


Figure 1: Use Case UML diagram

The interaction between the ARiS system and the Urban Planner/User is depicted in the use case diagram. The user can view the dashboard and accessibility map, select road segments, and analyse risk, compliance, and AI-based explanations. Filtering segments and switching view mode (rule-based and ML-based) support better data exploration.

Key features include the Scenario Simulator for modifying segment attributes and comparing results, and the Priority Intervention Engine for ranking segments based on urgency. The system also includes a City Planning Dashboard (Budget Optimizer) for resource allocation.

Additionally, users can predict accessibility risk for new areas, compare model outputs, and export PDF reports for Compliance and Scenario Simulator. The <<include>> relationships indicate dependencies between functions, ensuring a structured workflow.

Activity Diagram

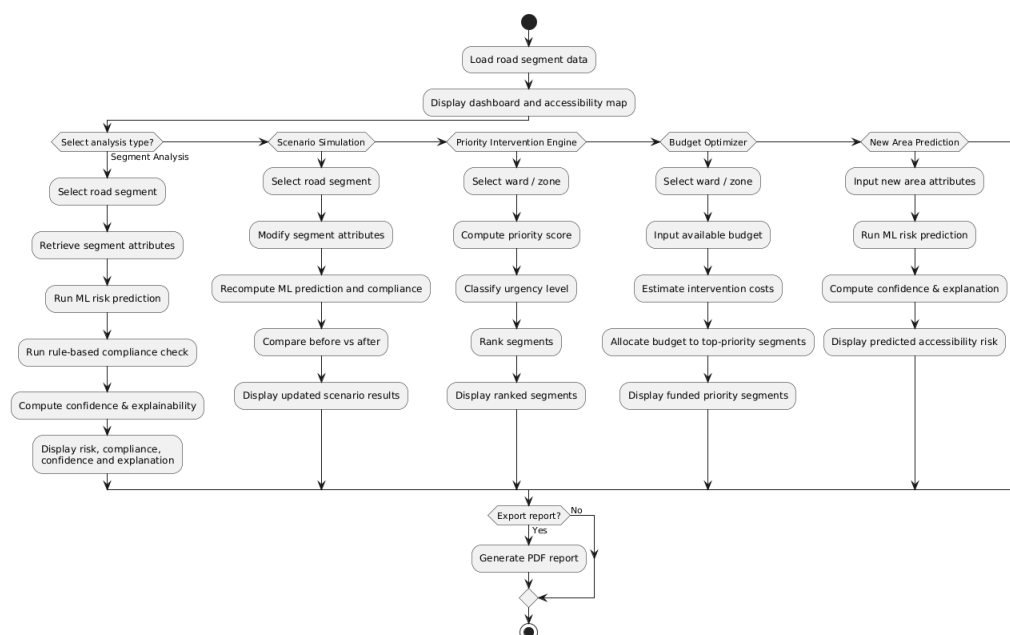


Figure 2: Activity UML diagram

The activity diagram represents the overall workflow of the ARiS System, beginning with loading road segment data and displaying the dashboard and accessibility map. The user then selects an analysis type such as Segment Analysis, Scenario Simulation, Priority Intervention engine, Budget optimization, or New Area Prediction. The system processes inputs based on the selection, performs ML prediction and compliance evaluation where required, and displays the results. Lastly, a PDF report export option is available to the user for Compliance and Scenario Simulator.

3.2 System Components

The system is made up of multiple components that work together to deliver its functionality. These components are divided into frontend modules, backend services, machine learning models, and supporting tools.

Frontend Components

The Frontend is responsible for presenting information to the user and enabling interaction with the system. React is utilized in its development, and Leaflet is used to render the map interface.

The central component is the map interface, where road segments are displayed as polylines. Each segment is colour-coded based on its risk level, allowing users to quickly identify high-risk areas. Users can click on individual segments to examine comprehensive data, such as compliance scores, predicted risk levels, and confidence values.

A sidebar panel provides additional functionality, including filtering options and comprehensive insights. By filtering segments according to risk levels or compliance status, users can more easily concentrate on particular areas of interest.

Together, these frontend components create an interactive and user-friendly environment for exploring accessibility data.

Backend Components

The backend is implemented using FastAPI and serves as the core processing unit of the system. It manages data flow, performs calculations, and handles communication between the frontend and the underlying models.

The backend provides several key APIs. Road segment data is provided by the data API in an organized manner. The prediction API generates machine learning predictions and confidence scores after processing input features. The feature importance API provides insights into which factors influence the model's predictions. Urgency scores are determined by the priority engine API, which then ranks segments accordingly. The scenario simulation API allows modification of segment attributes and recalculates predictions and compliance results. The budget optimizer API allocates available budget to top-priority segments based on estimated intervention costs. The new area prediction API enables prediction of accessibility risk for unseen areas using input attributes. All results are returned to the frontend dashboard for visualization and decision-making.

These APIs guarantee that the system operates efficiently and that all components remain well-integrated.

Machine Learning Model

A Random Forest classifier is used for predicting accessibility risk. This model was chosen because of its robustness in identifying complex patterns and its capacity to handle both numerical and categorical data.

The model is trained on a dataset containing key accessibility-related features, including sidewalk width, slope, curb ramp availability, traffic level, lighting quality, and surface condition. The model produces probabilities for each risk category (Low, Medium, High) during prediction. The category with the highest probability is selected as the predicted risk level.

Crucially, the probability value itself is used as a confidence score. This allows the system to indicate how certain the model is about its prediction, adding an extra layer of reliability.

Supporting Tools and Technologies

The system is built using a combination of modern technologies. Leaflet is utilized for map visualization and React and Tailwind CSS are used for frontend development. Python is the main programming language used in the development of the backend using FASTAPI.

Machine Learning functionality is implemented using Scikit-learn, while Pandas and NumPy are used for data processing. These tools provide a flexible and scalable foundation for the system.

3.3 Implementation Details

The system is implemented in a modular manner, ensuring that each component can operate independently while contributing to the overall functionality.

Data Preparation

```
id,name,lat_start,lng_start,lat_end,lng_end,sidewalk_width_m,slope_percent,has_curb_ramp,traffic_level,lighting_quality,surface_quality
block_1,Near CST Station,18.9398,72.8355,18.9485,72.8362,0.9,9,no,high,poor,broken
block_2,Marine Lines,18.9491,72.8277,18.9497,72.8285,1.3,5,yes,medium,good,smooth
block_3,Churchgate,18.9353,72.8302,18.9359,72.8310,1.8,3,yes,low,good,smooth
block_4,Kalbadevi Street,18.9490,72.8360,18.9496,72.8368,0.7,7,no,high,poor,broken
block_5,Fort Area Lane,18.9325,72.8350,18.9332,72.8357,1.1,6,no,medium,good,uneven
block_6,Hampankatta Link A,12.8704,74.8426,12.8711,74.8434,1.1,6,no,high,poor,uneven
block_7,Hampankatta Link B,12.8712,74.8436,12.8719,74.8444,1.5,3,yes,medium,good,smooth
block_8,Central Market Road A,12.8691,74.8468,12.8698,74.8476,0.9,7,no,high,poor,broken
block_9,Central Market Road B,12.8699,74.8478,12.8706,74.8486,1.2,5,no,medium,good,uneven
block_10,Clock Tower Approach A,12.8708,74.8490,12.8715,74.8498,1.6,3,yes,medium,good,smooth
```

Figure 3: Data Preparation code snippet 1

```
width = float(row["sidewalk_width_m"])
slope = float(row["slope_percent"])
has_ramp = str(row["has_curb_ramp"]).lower()
traffic = str(row["traffic_level"]).lower()
lighting = str(row["lighting_quality"]).lower()
surface = str(row["surface_quality"]).lower()
```

Figure 4: Data Preparation code snippet 2

```
if width < 0.9:
    violations.append("Severe: clear width extremely narrow (< 0.9m) [below CPWD minimum 1.2m]")
    score += 0.35
elif width < CPWD_MIN_WIDTH_M:
    violations.append("Non-compliant: clear width below CPWD minimum (1.2m)")
    score += 0.25
elif width < 1.5:
    violations.append("Below recommended width (~1.5m preferred for comfortable passing)")
    score += 0.10
```

Figure 5: Data Preparation code snippet 3

The input data consists of structured information about road segments. This data is pre-processed to ensure consistency and compatibility with both rule-based and machine learning components. Categorical variables are converted into numerical representations, allowing them to be used by the model.

Model Integration

```
# Numeric features
X = pd.DataFrame()
X["sidewalk_width_m"] = df["sidewalk_width_m"]
X["slope_percent"] = df["slope_percent"]
X["has_curb_ramp"] = df["has_curb_ramp"].str.lower().map({"yes": 1, "no": 0}).fillna(0)

# Categorical + fixed mapping (we control it)
traffic_map = {"low": 0, "medium": 1, "high": 2}
lighting_map = {"good": 1, "poor": 0}
surface_map = {"smooth": 0, "uneven": 1, "broken": 2}
```

Figure 6: Model Integration code snippet 1

```
if not MODEL_PATH.exists():
    print(f"[AURaMP] No ML model found at {MODEL_PATH}, using rule-based only.")
    return

bundle = joblib.load(MODEL_PATH)

ml_model = bundle.get("model")
risk_map_inv = bundle.get("risk_map_inv")
FEATURE_ORDER = bundle.get("meta", {}).get("features", [])
FEATURE_IMPORTANCES = bundle.get("meta", {}).get("feature_importances", [])
```

Figure 7: Model Integration code snippet 2

The Random Forest model that has been trained is stored and integrated into the backend. When a prediction request is received, the model processes the input features and returns probability scores for each risk category. The confidence score and the predicted risk level are then calculated using these probabilities.

Risk Prediction Workflow

```
def predict_risk_ml(block: dict):  
    """  
    Use ML model to predict risk_level for a given block.  
    Returns:  
    - risk_score (0-1 simplified severity index)  
    - risk_level ("low" / "medium" / "high")  
    - confidence (0-1, probability of predicted class)  
    - probs (list of class probabilities) or None  
    """  
    if ml_model is None or risk_map_inv is None:  
        # fallback: just return existing rule-based values, no confidence  
        return (  
            block.get("risk_score", 0.0),  
            block.get("risk_level", "unknown"),  
            None,  
            None,  
        )
```

Figure 8: Risk prediction code snippet 1

When a user selects a road segment, the system follows a defined workflow. The relevant features are extracted and passed to the backend. The machine learning model processes these features and return probability scores. The highest probability determines the predicted risk level, while the same value is used as the confidence score.

This process enables real-time predictions, allowing users to interact with the system dynamically.

Compliance Evaluation

```
// --- Rule-based risk (JS mirror of your backend rules) ---  
function computeRuleRisk(b) {  
    const violations = [];  
    let score = 0;  
  
    const width = Number(b.sidewalk_width_m);  
    const slope = Number(b.slope_percent);  
    const hasRamp = String(b.has_curb_ramp).toLowerCase();  
    const traffic = String(b.traffic_level).toLowerCase();  
    const lighting = String(b.lighting_quality).toLowerCase();  
    const surface = String(b.surface_quality).toLowerCase();
```

Figure 9: Compliance Evaluation code snippet 1

The rule-based engine evaluates each segment against predefined criteria. Individual checks are combined to produce a compliance score, expressed as a percentage. Segments that don't meet important standards are labelled as non-compliant.

This evaluation provides a clear and interpretable assessment of accessibility conditions.

Scenario Simulation

```
score, level, confidence, probs = predict_risk_ml(block)  
explanation = explain_risk_ml(block)  
  
probs_dict = None  
if probs is not None and len(probs) >= 3:  
    probs_dict = {  
        "low": float(probs[0]),  
        "medium": float(probs[1]),  
        "high": float(probs[2]),  
    }
```

Figure 10: Scenario Simulation code snippet 1

```
const before = useMemo(() => {  
    if (!selectedBlock) return null;  
    const comp = computeCompliance(selectedBlock);  
    const risk = computeRuleRisk(selectedBlock);  
    return { comp, risk };  
}, [selectedBlock]);  
  
const [mlScenarioResult, setMlScenarioResult] = useState(null);  
  
// Scenario -> call /predict-ml with scenarioBlock  
useEffect(() => {  
    if (!scenarioBlock) return;
```

Figure 11: Scenario Simulation code snippet 2

The scenario simulator allows users to explore “what-if” situations by modifying segment attributes. When modifications are performed, the updated data is sent to the backend for recalculation of both machine learning and rule-based evaluations.

The results are then displayed instantly, enabling users to understand how specific improvements can affect accessibility outcomes.

Model Comparison

```
const ModelComparison = ({ blocks = [], onSelectBlock }) => {
  const comparison = useMemo(() => {
    const validBlocks = blocks.filter((b) => b.risk_level && b.ml_risk_level);

    const agreement = validBlocks.filter(
      (b) => b.risk_level === b.ml_risk_level
    );

    const disagreement = validBlocks.filter(
      (b) => b.risk_level !== b.ml_risk_level
    );

    const matrix = {
      high: { high: 0, medium: 0, low: 0 },
      medium: { high: 0, medium: 0, low: 0 },
      low: { high: 0, medium: 0, low: 0 },
    };
  });
};
```

Figure 12: Model Comparison code snippet 1

The machine learning model and the rule-based engine's outputs are compared by the system. Both risk levels are examined for each segment to determine whether they match or differ. This comparison helps validate the model and provides insight into its behaviour.

Confidence Layer Implementation

```
for block in blocks:
    ml_score, ml_level, ml_conf, probs = predict_risk_ml(block)

    probs_dict = None
    if probs is not None and len(probs) >= 3:
        probs_dict = {
            "low": float(probs[0]),
            "medium": float(probs[1]),
            "high": float(probs[2]),
        }
```

Figure 13: Confidence Layer code snippet 1

```
if ml_model is None or risk_map_inv is None:
    # fallback: just return existing rule-based values, no confidence
    return {
        block.get("risk_score", 0.0),
        block.get("risk_level", "unknown"),
        None,
        None,
    }

features = encode_features(block)
X = [features] # shape (1, n_features)
```

Figure 14: Confidence Layer code snippet 2

The confidence layer is derived from the probability output of the machine learning model. The highest probability value is used as the confidence score, indicating how strongly the model supports its prediction.

Lower confidence values suggest uncertainty, while higher values indicate more reliable predictions. This information is provided to users to support better decision-making.

Priority Ranking Process

```
# compliance
if seg.compliance_score < 50:
    factors.append("Low compliance score")
elif seg.compliance_score < 70:
    factors.append("Moderate compliance deficiency")

# width
if seg.sidewalk_width is not None:
    if seg.sidewalk_width < 1.0:
        factors.append("Very narrow sidewalk width")
    elif seg.sidewalk_width < 1.5:
        factors.append("Below-recommended sidewalk width")
```

Figure 15: Priority Ranking code snippet 1

```
if not payload.segments:
    raise HTTPException(status_code=400, detail="No segments provided.")

out_segments: List[SegmentOut] = []
risk_values: List[float] = []
comp_values: List[float] = []

for seg in payload.segments:
    score = compute_priority_score(seg)
    level = classify_priority_level(score)
    explanation = build_priority_explanation(seg)
    actions = build_recommended_actions(seg, level)
```

Figure 16: Priority Ranking code snippet 2

The priority engine processes each segment by extracting relevant attributes and converting them into scoring components. These components are combined using a weighted approach to produce a final priority score.

After that, segments are then classified into priority levels and sorted accordingly. This ranking gives a clear picture of which areas require immediate attention.

Accessibility Risk Prediction for New Areas

```
const handleDraw = (e) => {
  const layer = e.layer;

  setTempGeometry(layer); // store drawn geometry
  setShowPredictionForm(true);
};
```

Figure 17: Accessibility Risk Prediction code snippet 1

```
# base cost by priority level
if level == "Critical":
    cost = 250000.0
elif level == "High":
    cost = 180000.0
elif level == "Medium":
    cost = 100000.0
else:
    cost = 60000.0
```

Figure 18: Accessibility Risk Prediction code snippet 2

The system includes a feature for predicting accessibility risk in newly defined or previously unassessed areas. Users can draw or define a new road segment and provide relevant input features such as sidewalk width, slope, curb ramp, traffic level, lighting quality, and surface condition. The backend then processes these values, generating a confidence score and predicting the risk level using the trained machine learning model.

This feature extends the system beyond analysis of existing dataset entries and allows it to support future planning scenarios. It is particularly useful for evaluating proposed infrastructure changes or newly developing urban areas before implementation.

Model Insights (Global Feature Importance)

```
features = []
for name, raw_value, norm_val in zip(FEATURE_ORDER, importances, norm_importances):
    features.append(
        {
            "name": name,
            "raw_importance": float(raw_value),
            "normalized_importance": float(norm_val),
        }
    )
```

Figure 19: Model insights code snippet

The system provides model insights using global feature importance. It identifies which input features have the highest influence on risk prediction. This helps in understanding overall model behaviour.

Explainability (XAI) for Predictions

```
const ExplanationBars = ({ explanation, prettyFeatureName }) => {
  if (!explanation || explanation.length === 0) return null;

  // Find the largest absolute impact so we can normalize everything to 0-100%
  const maxImpact =
    Math.max(...explanation.map((e) => Math.abs(e.impact))) || 1;
}
```

Figure 20: Explanation for ML Predictions

The system includes an explainability module to understand ML predictions. It highlights key features influencing each risk level for a segment. This improves transparency and supports better understanding of model decisions.

City Planning Dashboard (Budget Optimizer)

```
@app.post("/budget-optimizer", response_model=BudgetResponse)
def run_budget_optimizer(payload: BudgetRequest) -> BudgetResponse:
    """
    Select the best segments that can be funded within the available budget.
    Uses greedy selection based on descending priority score.
    """
    if not payload.segments:
        raise HTTPException(status_code=400, detail="No segments provided.")

    ranked_segments = []

    # First compute priority for all segments
    for seg in payload.segments:
        score = compute_priority_score(seg)
        level = classify_priority_level(score)
        explanation = build_priority_explanation(seg)
        actions = build_recommended_actions(seg, level)
        estimated_cost = estimate_intervention_cost(seg, level)
```

Figure 21: City Planning code snippet 1

```
const NewSegmentModal = ({ onSubmit, onClose }) => {
  const [formData, setFormData] = useState({
    sidewalk_width_m: 1.2,
    slope_percent: 5,
    has_curb_ramp: "yes",
    traffic_level: "medium",
    lighting_quality: "good",
    surface_quality: "smooth",
  });
```

Figure 22: City Planning code snippet 2

The system also has a City Planning Dashboard with a budget optimization feature to support planning and resource allocation. After the priority intervention engine ranks segments based on urgency, the budget optimizer estimates intervention costs and selects the set of segments that can be addressed within a given budget.

The backend calculates estimated costs based on the severity and type of accessibility issues, while the optimizer allocates available resources to the highest-priority segments first. The final output includes the funded segments, budget used, and remaining balance.

This feature improves the practical value of the system by linking accessibility analysis with financial decision-making, making it more relevant for real urban planning applications.

System Integration and Visualisation

REST APIs allow communication between the frontend and backend, allowing smooth data flow between components. The final results are visualised on an interactive map deployed in vercel, where users can explore accessibility conditions in detail.

Colour-coded segments, detailed pop-ups, and filtering options provide a clear and intuitive representation of the data.

Chapter 4: Performance Evaluation Setup and Scenarios

4.1 Evaluation Methodology

The evaluation of the proposed system focuses on understanding how effectively it identifies accessibility risks and supports decision-making in a practical context. Instead of relying only on traditional machine learning metrics, the evaluation considers how well the system behaves across realistic scenarios and whether the outputs are logically consistent and useful for urban planning.

One of the main aspects evaluated is prediction consistency. The system is expected to produce results that align with real-world understanding of accessibility. For example, road segments with conditions such as narrow sidewalks, poor lighting, and high traffic are more likely to be classified as high risk. The model's ability to consistently reflect such patterns, considering all relevant infrastructure features, is used as a key indicator of performance.

The confidence score generated by the machine learning model is another important metric. For every risk category, the model generates probability values, and the highest probability represents how confident the model is in its prediction. A higher confidence

value indicates stronger agreement within the model, while a lower value suggests uncertainty. This is particularly useful in identifying cases where predictions may require further validation.

Additionally, the system also evaluates agreement between rule-based and machine learning outputs. For each road segment, both approaches generate a risk level, and these results are compared. When both outputs match, it indicates strong alignment between predefined standards using CPWD guidelines (India's official guidelines) and data-driven predictions. When they differ, it highlights areas where the model may be capturing patterns that are not explicitly defined in rules. This comparison helps improve transparency and provides additional insight into system behaviour.

Another important metric used in evaluation is the compliance score. Based on predefined thresholds, this score indicates how well a segment satisfies accessibility requirements. It provides a clear and interpretable baseline, which can be used to assess whether machine learning predictions are reasonable.

Ranking effectiveness is the main focus of the priority intervention engine. Segments identified as high priority should generally correspond to those with higher risk and lower compliance. The system's ability to correctly rank such segments is used to evaluate how useful it is for planning and resource allocation.

Overall, the evaluation approach combines prediction consistency, confidence, agreement, and practical relevance. This ensures that the system is both technically functional and useful in real-world applications.

4.2 Test Environment

The system was developed and tested in a local environment using standard hardware and widely used software tools. The goal was to maintain real-time responsiveness while ensuring smooth integration across various components.

The frontend application was built using React and executed in a web browser. It offers an interactive interface that allows users to examine the map and evaluate accessibility information. The backend was developed using FastAPI, and a Python server was used to run it locally. HTTP requests allow communication between the frontend and the backend, enabling effective data transmission.

The Scikit-learn library was utilized to implement the machine learning model. During runtime, it is loaded into the backend and used to generate predictions whenever required. Pandas and NumPy are used for data processing operations including feature encoding and transformation.

The system was tested on a standard personal computer, which was sufficient to manage backend processing and frontend rendering at the same time. External servers or specialised hardware were not required. This demonstrates that the system is lightweight and can be deployed without significant infrastructure requirements.

The map interface was rendered using the Leaflet library for visualization. It enables smooth interaction with geospatial data, including zooming, panning, and selecting road segments. All components were integrated within a single development setup, ensuring consistency during testing.

This environment provided a stable and controlled setup for evaluating system performance across different scenarios.

4.3 Test Scenarios

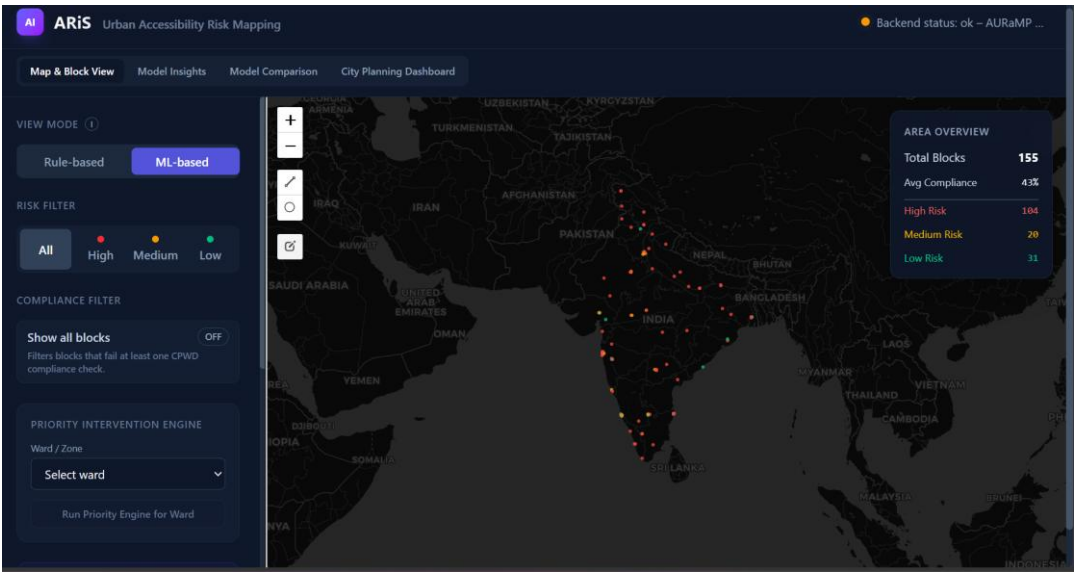


Figure 23: ARiS

A variety of test scenarios were created to assess the system’s performance in various accessibility conditions. These scenarios simulate real-world situations by varying input attributes and observing how the system responds.

The first set of scenarios focused on high-risk conditions. These included segments with narrow sidewalks, poor lighting, high traffic levels, and missing curb ramps. Such conditions are strong indicators of accessibility issues, and in most cases, the system classified these segments as high risk. The predictions are based on the combined influence of all relevant features. The confidence scores were also relatively high, indicating that the model was certain about its predictions. The priority intervention engine ranked these segments as critical, which aligns with expected behaviour.

The second set of scenarios represented low-risk conditions. The segments had wider sidewalks, good lighting, low traffic, and appropriate infrastructure. The system correctly identified these as low-risk areas, with high compliance scores and low priority levels. The classification is based on combined influence of all relevant features. This demonstrates that the system can effectively recognise safe and accessible environments.

The third set of scenarios involved mixed or moderate conditions, where some attributes met accessibility standards while others did not. For example, a segment might have sufficient sidewalk width but poor lighting. In such cases, the system typically produced medium risk classifications. Confidence scores were also moderate, reflecting some level of uncertainty. This shows that the system can handle more complex and ambiguous situations.

The scenario simulation feature was also tested by modifying attributes in real time. For example, risk levels and compliance scores changed noticeably when sidewalk width was

increased or lighting was improved. This confirms that the system responds dynamically to input changes and can effectively simulate potential improvements.

The model comparison module was evaluated by analysing cases where rule-based and machine learning outputs differed. These differences highlight situations where the model identifies patterns beyond predefined rules, demonstrating the added value of machine learning in capturing complex relationships.

The system was also tested on new or user-defined inputs representing infrastructure conditions. It successfully generated risk predictions, confidence scores, and explanations, demonstrating its ability to generalise to unseen areas.

The model insights module was evaluated using global feature importance. The system identifies the most influential features in risk prediction. Sidewalk width and lighting showed higher impact, followed by other factors. This confirms the model captures meaningful feature relationships.

The explainability (XAI) module was tested to understand individual predictions. The system highlights key contributing features for each segment. For example, poor lighting or narrow width showed higher impact in high-risk cases. This improves transparency and understandability.

A scenario with a limited budget was used to evaluate the city planning dashboard. Based on priority scores, the optimizer selected segments that maximised accessibility improvement, demonstrating effective support for resource allocation and planning decisions.

Finally, the confidence layer was assessed by observing how probability values change across different scenarios. High confidence was observed in clearly defined situations, while lower confidence appeared in more ambiguous cases. This behaviour is expected and supports the reliability of the system.

Overall, the test scenarios show that the system performs consistently across a range of conditions. It accurately identifies risk levels, reflects compliance status, and provides meaningful insights through prioritisation and simulation.

Chapter 5: Results and Discussions

5.1 Experimental Results

Test Case 1 (System Startup)

Field	Details
Test Case ID	TC001
Test Case Name	Load Dashboard Successfully
Module/Feature	System Initialization
Preconditions	Backend is running.
Test Steps	1. Launch Application. 2. Observe dashboard.
Test Data	Valid backend connection.
Expected Result	Dashboard loads with map and data.
Actual Result	Loaded successfully.
Status	Pass

Test Case 2 (Map Visualization)

Field	Details
Test Case ID	TC002
Test Case Name	Display Road Segments on Map
Module/Feature	Map Visualization
Preconditions	Dataset available
Test Steps	1. Open App. 2. Check Map.
Test Data	Road segment dataset.
Expected Result	Segments displayed on map.
Actual Result	Displayed correctly.
Status	Pass

Test Case 3 (User Interaction)

Field	Details
Test Case ID	TC003
Test Case Name	Select Segment from Map
Module/Feature	Map Interaction
Preconditions	Segments visible
Test Steps	1. Click segment. 2. Observe details.
Test Data	Any segment.
Expected Result	Segment details displayed.
Actual Result	Working correctly.
Status	Pass

Test Case 4 (Filtering Logic)

Field	Details
Test Case ID	TC004
Test Case Name	Risk Filtering Functionality
Module/Feature	Risk Filter
Preconditions	Mixed risk data
Test Steps	1. Select High/Medium/Low filter. 2. Observe map.
Test Data	Segments with different risks.
Expected Result	Correct filtering applied.
Actual Result	Filtering works correctly.
Status	Pass

Test Case 5 (Compliance Check)

Field	Details
Test Case ID	TC005
Test Case Name	Non-Compliant Filter
Module/Feature	Compliance Filter
Preconditions	Compliance data available
Test Steps	1. Enable Filter. 2. Observe results.
Test Data	Mixed compliance segments.
Expected Result	Only non-compliant shown.
Actual Result	Works correctly.
Status	Pass

Test Case 6 (Core Feature)

Field	Details
Test Case ID	TC006
Test Case Name	ML Risk Prediction for Selected Segment
Module/Feature	ML Prediction
Preconditions	ML model is loaded.
Test Steps	1. Select a road segment. 2. Observe ML risk output.
Test Data	Segment with valid features.
Expected Result	ML risk score and level are displayed correctly.
Actual Result	ML prediction displayed successfully.
Status	Pass

Test Case 7 (XAI)

Field	Details
Test Case ID	TC007
Test Case Name	XAI Explanation for Selected Segment
Module/Feature	Explainability Layer
Preconditions	ML prediction is available.
Test Steps	1. Select a segment. 2. View explanation panel.
Test Data	Segment with ML explanation.
Expected Result	Key features influencing risk are displayed.
Actual Result	Explanation shown correctly.
Status	Pass

Test Case 8 (Simulator)

Field	Details
Test Case ID	TC008
Test Case Name	Scenario Simulator Updates Risk Output
Module/Feature	Scenario Simulator
Preconditions	Segment selected.
Test Steps	1. Modify values (width, slope, curb ramp, traffic, surface, and lighting). 2. Observe updated results.
Test Data	Modified segment values.
Expected Result	Risk and compliance update dynamically.
Actual Result	Outputs updated correctly.
Status	Pass

Test Case 9 (Priority Navigation)

Field	Details
Test Case ID	TC009
Test Case Name	Accessibility Risk Prediction for New Area
Module/Feature	New Area Risk Prediction
Preconditions	Backend ML model is loaded and running
Test Steps	1. Open the application. 2. Navigate to new area prediction / drawing feature 3. Input attributes (e.g., sidewalk width, slope, curb ramp, traffic, lighting, surface). 4. Submit the input. 5. Observe predicted output.
Test Data	New segment input values.
Expected Result	- System predicts risk level. - Displays risk score. - Shows confidence value. - Provides explanation
Actual Result	Risk prediction, confidence, and explanation displayed correctly.
Status	Pass

Test Case 10 (Planning feature)

Field	Details
Test Case ID	TC010
Test Case Name	Priority Engine Ranks Segments
Module/Feature	Priority Intervention Engine
Preconditions	Ward selected.
Test Steps	1. Select ward. 2. Run engine. 3. Observe ranking.
Test Data	Ward-based segment data.
Expected Result	Segments ranked by urgency.
Actual Result	Ranking generated correctly.
Status	Pass

Test Case 11 (Confidence)

Field	Details
Test Case ID	TC011
Test Case Name	Confidence Score Display
Module/Feature	Uncertainty & Confidence Layer
Preconditions	ML prediction available.
Test Steps	1. Select segment. 2. Observe confidence score.
Test Data	ML probability output.
Expected Result	Confidence value displayed correctly.
Actual Result	Confidence shown correctly.
Status	Pass

5.2 Performance Analysis

The performance of the system is analysed based on how well the results align with expected real-world behaviour.

Strong alignment with expected patterns was shown by the machine learning model. High-risk predictions were consistently associated with poor infrastructure conditions, while low-risk predictions corresponded to well-designed segments. This indicates that the relationship between input features and accessibility risk has been successfully learned by the model.



Figure 24: confidence score snippet 1



Figure 25: confidence score snippet 2

The model's performance is further supported by the confidence scores. Higher confidence values were observed in scenarios where input conditions clearly indicated a specific risk level. On the other hand, lower confidence values appeared in mixed or borderline cases.

This behaviour reflects appropriate uncertainty handling and improves the reliability of the system.

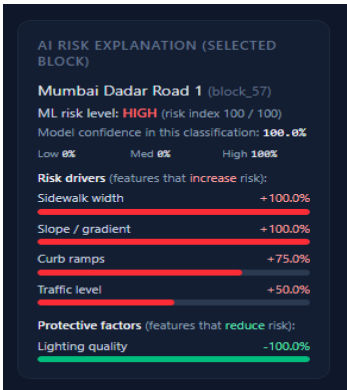


Figure 26: AI risk explanation

The explainability (XAI) module performed effectively in understanding predictions. The system highlighted key contributing features for each segment. This improves transparency and supports better understanding of model decisions.



Figure 27: Compliance snippet 1



Figure 28: Compliance snippet 2

The rule-based compliance engine provided a consistent baseline for evaluation. It produced predictable results that met established accessibility guidelines. When compared with machine learning predictions, a high level of agreement was observed in straightforward scenarios. This confirms that the model is consistent with established rules in most cases.

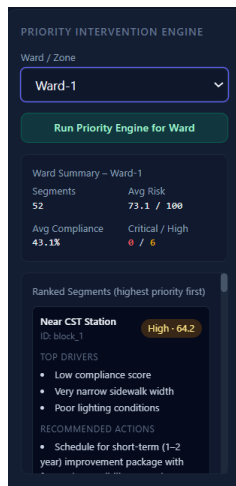


Figure 29: Priority Engine snippet 1

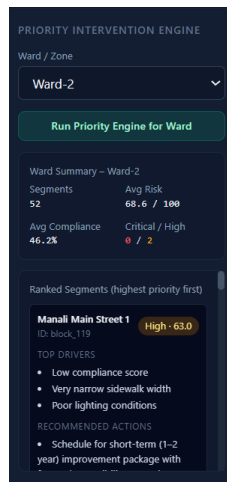


Figure 30: Priority Engine snippet 2

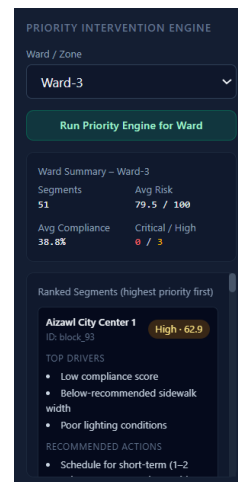


Figure 31: Priority Engine snippet 3

The priority intervention engine performed effectively in ranking segments. Segments with higher risk and lower compliance were consistently assigned higher priority levels. This indicates that the scoring mechanism successfully captures the urgency of intervention.

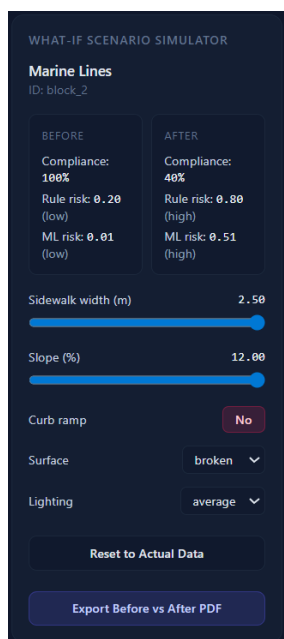


Figure 32: Scenario Simulator snippet 1

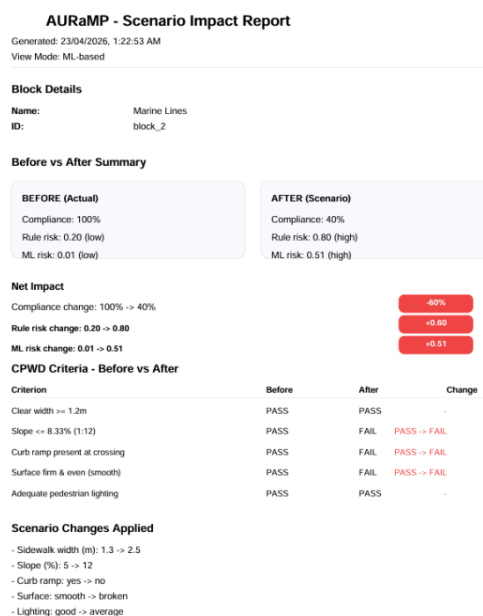


Figure 33: Scenario Simulator snippet 2

Strong responsiveness was shown by the scenario simulation feature. Changes in input parameters resulted in immediate and logical changes in system outputs. This confirms that the system can be used as a practical tool for evaluating potential improvements.

The system worked well in terms of usability, producing results that were understandable and straightforward. The combination of risk levels, compliance scores, confidence values, and priority rankings allows users to understand accessibility conditions from multiple perspectives.

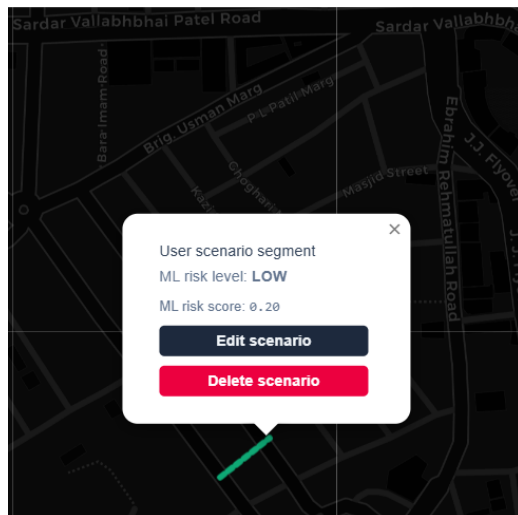


Figure 34: Accessibility Risk Prediction for New Area snippet 1

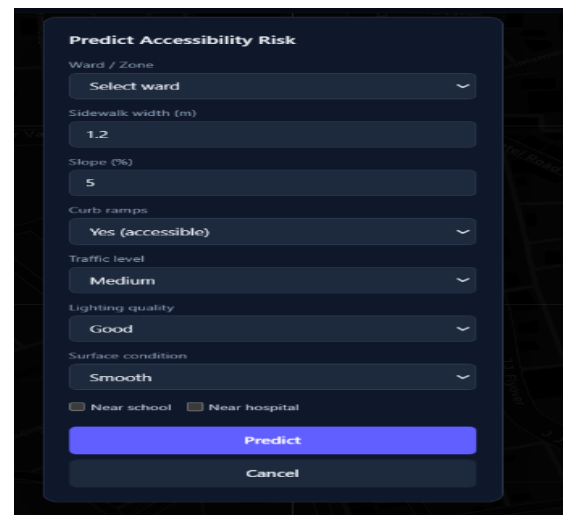


Figure 35: Accessibility Risk Prediction for New Area snippet 2

The system also performed well in predicting accessibility risk for new or user-defined inputs. It generated consistent risk levels, confidence scores, and explanations without relying on predefined data, demonstrating its ability to generalise to unseen scenarios.

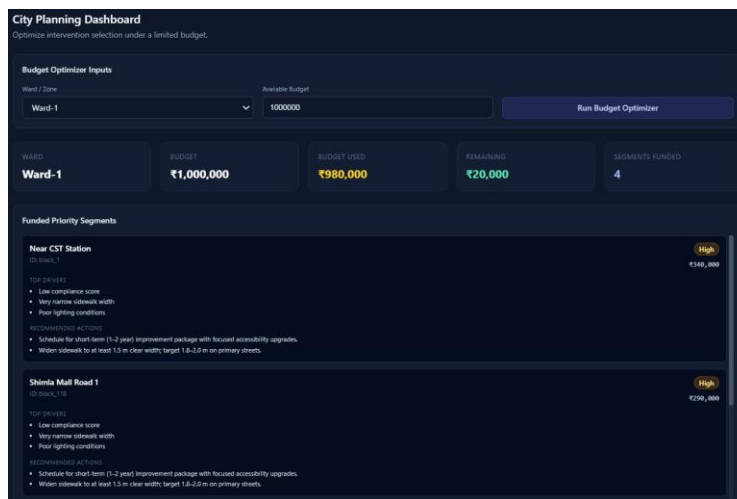


Figure 36: Budget Optimizer

The city planning dashboard, which is the budget optimizer, performed effectively by prioritising segments within budget constraints. Based on priority scores, it selected segments with higher impact, supporting practical decision-making and resource allocation.

Overall, the system shows strong performance in both prediction accuracy and practical applicability. The results align well with expected outcomes, indicating that the system is reliable and effective for its intended purpose.

5.3 Discussion

The results highlight several important observations about the system's behaviour and effectiveness.

One key observation is the strong relationship between infrastructure quality and predicted risk. Segments with multiple accessibility issues were consistently classified as high risk,

while well-designed segments were classified as low risk. This confirms that the system is correctly capturing the impact of key features on accessibility.

Another important observation is the role of the confidence layer. By providing a measure of prediction reliability, the system allows users to distinguish between strong and uncertain predictions. This is particularly useful in borderline cases, where additional validation may be required.

The comparison between rule-based and machine learning outputs provides valuable insight into system behaviour. In many cases, both approaches produced similar results, reinforcing the validity of the predictions. However, in more complicated situations, differences were noted. These differences indicate that the machine learning model can identify relationships that aren't specifically defined in rules, adding depth to the analysis.

Another observation is the role of the explainability (XAI) module. It highlights key features influencing each prediction. This improves transparency and understanding of model outputs.

The priority intervention engine turned out to be a helpful tool for making decisions. By combining multiple factors into a single score, the system is able to highlight areas that require immediate attention. This simplifies the process of identifying critical segments and supports efficient resource allocation.

The system's practical value is further increased by the scenario simulation capability. It allows users to explore the impact of potential improvements, providing a predictive view of how changes can affect accessibility. As a result, the system is both exploratory and analytical.

The system's ability to predict accessibility risk for new inputs further strengthens its practical applicability. It demonstrates that the model can generalise beyond predefined data and handle previously unseen scenarios effectively.

The city planning dashboard, including the budget optimizer, also enhances the system's decision-support capability. By selecting high-impact segments under budget constraints, it helps translate analytical results into actionable planning strategies.

Despite these advantages, many drawbacks were noted. The machine learning model depends on the quality and diversity of the training data. The model's prediction might be less accurate if the data does not accurately reflect real-world conditions. Furthermore, the rule-based system is limited by predefined thresholds, which may not capture all possible scenarios.

Another drawback is that the system now runs in a controlled environment with predefined or simulated data. Real-world deployment would require integration with live data sources, such as GIS datasets or sensor inputs.

Overall, the system demonstrates strong potential as a decision-support tool, while also highlighting areas for future improvement.

Chapter 6: Conclusions and Future Works

6.1 Summary of Contributions

This project presented the development of an AI-Driven Accessibility Risk Mapping and Prediction Intelligence System (ARiS), designed to assess accessibility conditions at a detailed, road-segment level. The system successfully integrates machine learning, rule-based evaluation, and interactive visualisation to support data-driven urban planning.

The primary objective of the project was to create a system capable of identifying accessibility risks based on key environmental and infrastructure-related factors. This objective has been achieved through the implementation of a machine learning model that predicts risks levels using features such as sidewalk width, slope, curb ramp, traffic conditions, lighting, and surface quality. The model provides not only predictions but also confidence scores, allowing users to understand the reliability of each output.

In addition to prediction, the project introduced a rule-based compliance engine that evaluates accessibility against predefined standards. This ensures that the system remains interpretable and aligned with established guidelines. The integration of both approaches provides a balanced solution that combines transparency with predictive capability.

Another key contribution is the development of a scenario simulation module. This feature enables users to modify input parameters and observe how changes impact accessibility outcomes. As a result, the system goes beyond static analysis and supports exploratory decision-making.

The project also included a priority intervention engine, which ranks road segments based on urgency. By combining risk, compliance, and contextual factors, the system helps identify areas that require immediate attention. This directly supports efficient resource allocation in urban planning.

Moreover, system transparency is improved by adding a model comparison module and confidence layer. These features allow users to assess prediction reliability and understand differences between rule-based and machine learning outputs.

Overall, the project achieves its goal of developing a comprehensive and practical tool for accessibility assessment. It demonstrates how artificial intelligence can be applied to real-world urban challenges in a meaningful and interpretable way.

6.2 Limitations

While the system demonstrates strong functionality, several limitations were identified during development and testing.

The dataset used to train the machine learning model is one of the primary drawbacks. The model's performance is directly dependent on the quality and diversity of the data. The predictions might not generalize well to all situations if the dataset does not accurately represent real world variability.

The rule-based compliance engine is also limited by predefined thresholds. While these rules provide clarity and interpretability, they may not capture all possible accessibility conditions. Real-world environments can be more complex than fixed criteria allow.

The usage of a controlled testing environment is another drawback. The system was developed and evaluated using predefined or simulated data. As a result, real-world deployment may introduce additional challenges, such as incomplete data, noise, or inconsistencies in data sources.

The machine learning model, although effective, may produce uncertain predictions in borderline cases. While the confidence layer helps identify such situations, it does not eliminate the need for human validation in critical decision-making scenarios.

The current implementation focuses primarily on a limited set of features, such as sidewalk width, lighting, and traffic conditions, in terms of system scope. Other relevant factors, such as pedestrian density, weather conditions, or temporal variations, are not included.

Finally, the system currently operates as a standalone application. It is not yet integrated with external urban planning systems or real-time data sources, which may limit its practical deployment in large-scale environments.

6.3 Future Work

Several opportunities exist to extend and improve the system in future work.

One important direction is the integration of real-world data sources, such as Geographic Information Systems (GIS), sensor data, or open urban datasets. This would enable the system to function with real-time data and deliver more precise and current insights.

The machine learning model can also be enhanced by incorporating larger and more diverse datasets, improving its ability to generalise across different urban environments. Additionally, advanced techniques such as model calibration or ensemble tuning could be applied to improve prediction reliability.

The expansion of input features is another area for improvement. Including additional factors such as pedestrian flow, weather conditions, and time-based variations would provide a more comprehensive assessment of accessibility.

The system could also be extended to include confidence-aware decision-making, where priority rankings are adjusted based on prediction certainty. This would further improve the reliability of the priority intervention engine.

From a usability perspective, the system can be enhanced by developing a more advanced dashboard interface, including richer visualisations, reporting features, and integration with planning tools.

Another crucial factor is scalability. Future work could focus on deploying the system in a cloud-based environment, enabling it to handle larger datasets and support multiple users simultaneously.

Finally, the system could be adapted for use in different geographic regions, considering local accessibility standards and regulations. This would increase its applicability and impact.

REFERENCES

1. Batty, M., Axhausen, K.W., Giannotti, F., Pozdnoukhov, A., Bazzani, A., Wachowicz, M., Ouzounis, G. and Portugali, Y. (2012). Smart cities of the future. *The European Physical Journal Special Topics*, [online] 214(1), pp.481–518. Available at: <https://doi.org/10.1140/epjst/e2012-01703-3>.
2. Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), pp.5–32. [online] Available at: <https://link.springer.com/article/10.1023/A:1010933404324>.
3. Doshi-Velez, F. and Kim, B. (2017). *Towards A Rigorous Science of Interpretable Machine Learning*. [online] Available at: <https://arxiv.org/pdf/1702.08608>.
4. esri (2022). *Esri*. [online] Esri.com. Available at: <https://www.esri.com/en-us/arcgis/geospatial-platform/overview?rsource=https%3A%2F%2Fwww.esri.com%2Fen-us%2Farcgis%2Fabout-arcgis%2Foverview>.
5. Google (2019). *Accessibility in Google Maps - Computer - Google Maps Help*. [online] Google.com. Available at: https://support.google.com/maps/answer/6396990?visit_id=639124360754881900-2177981835&rd=1.
6. Imrie, R. and Hall, P. (2003). *Inclusive Design*. Taylor & Francis. [online] Available at: <https://doi.org/10.4324/9780203362501>.
7. India, G. of (2024). *Smart Cities Mission Achievements*. [online] Pib.gov.in. Available at: <https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2085711@=3&lang=2>
8. Lundberg, S., Allen, P. and Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. [online] Available at: <https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf>.
9. Saaty, T. (2008). Decision making with the analytic hierarchy process. *Int. J. Services Sciences*, [online] 1(1), pp.83–98. Available at: <https://www.rafikulislam.com/uploads/resourses/197245512559a37aadea6d.pdf>.
10. StreetLight (2018). *Transportation Analytics On Demand | StreetLight Data*. [online] StreetLight Data. Available at: <https://www.streetlightdata.com/>.
11. SUSAN HANDY, R.E. (2026). *Welcome To Zscaler Directory Authentication*. [online] Scispace.com. Available at: <https://scispace.com/papers/measuring-the-unmeasurable-urban-design-qualities-related-to-1ieq5q5prc>.
12. Yigitcanlar, T., Kamruzzaman, Md., Foth, M., Sabatini-Marques, J., da Costa, E. and Ioppolo, G. (2019). Can cities become smart without being sustainable? A systematic review of the literature. *Sustainable Cities and Society*, 45, pp.348–365. <https://doi.org/10.1016/j.scs.2018.11.033>.
13. Zheng, Y. (2015). Trajectory Data Mining. *ACM Transactions on Intelligent Systems and Technology*, 6(3), pp.1–41. <https://doi.org/10.1145/2743025>.
14. DISABILITY RIGHTS ALLIANCE (India). (2021). *HARMONISED GUIDELINES*. [online] Available at: <https://disabilityrightsallianceindia.wordpress.com/about/resources/standards-guidelines/harmonised-guidelines/>

APPENDIX

Ethics Form

Research Ethics Screening Form for Students

Middlesex University is concerned with protecting the rights, health, safety, dignity, and privacy of its research participants. It is also concerned with protecting the health, safety, rights, and academic freedom of its students and with safeguarding its own reputation for conducting high quality, ethical research.

This Research Ethics Screening Form will enable students to self-assess and determine whether the research requires ethical review and approval before commencing the study.

Student Name: Rayaan Rafique		Email: RR1080@live.mdx.ac.uk
	Research project title: AI-Driven Urban Accessibility Risk Mapping and Prediction System	
	Programme of study/module: BENG – Computer Systems Engineering	
Supervisor Name: Mr. Roshan Renji		Email: R.Renji@mdx.ac.uk

Please answer the following questions to determine whether your proposed activity requires ethical review and approval

1. Will the research 'involve human participants,' with or without their knowledge or consent? <i>('Human participants' is a wide phrase including, but not limited to, observation, questionnaires, interviews (online and hard-copy), focus groups, social media platforms, etc., visual recordings (e.g., photos, video), audio recordings (e.g., digital, tape), or other human data/materials (e.g., blood, saliva, tissues or other human samples). It also includes yourself in cases where you, as the researcher, are planning to conduct research on yourself or to be involved in the same way as other participants in the project)</i>	☐ Yes	☒ No
2. Will the research involve animals or animal parts?	☐ Yes	☒ No
3. Will the research involve any activity that might cause damage or present a significant risk to society? (e.g., to precious artefacts or the environment)	☐ Yes	☒ No
4. Is the research likely to put you or others to any risks other than considered everyday risks? (e.g., risk of physical or psychological harm, engagement in illegal activities, working in a foreign country, travel risks, working alone, breaching security systems or searching the Internet for data about highly sensitive topics such as sexual abuse, terrorism, etc.)	☐ Yes	☒ No
5. Will the research include digital information/data from the internet, social media platforms, Apps, or smart devices with or without users' knowledge or consent, and/or could it lead to users being identified?	☐ Yes	☒ No
6. Will the research require approval to access any data? (e.g., access data through individuals and/or data through an external organisation(s))	☐ Yes	☒ No
7. Could anyone involved in the research have a potential conflict of interest or lack of impartiality?	☐ Yes	☒ No
8. Will your project involve working with any substances and/or equipment that may be considered hazardous to you or others?	☐ Yes	☒ No

9. Will the research involve discussion of sensitive topics? (e.g., sexual activity, drug use, national security etc.)	☐	☒
	Yes	No
10. Will the outputs from your research (e.g., products, reports, publications, etc.) likely cause any harm to you, others, or to society; or have legal issues?	☐	☒
	Yes	No

If you have answered 'Yes' to ANY of the above questions, your application requires ethical review and approval prior to commencing your research. Please complete the 'Application for Ethical Approval for Research Projects for Students' form

If you have answered 'No' to ALL of the above questions, your application may not require ethical review and approval before commencing your research. Your research supervisor will confirm this below.

Student Signature: Rayaan Rafique Date: 2/2/2026

To be completed by the supervisor:

Based on the details provided in the self-assessment form, I confirm that:	
The study does not require ethical review and approval	☒
The study requires ethical review and approval	

Supervisor Signature:.....RR...(Roshan Renji)..... Date:....22/6/2026.....

Generative AI Use Disclaimer Form

Generative AI Use Disclaimer Form

This form is intended for PDE3823 students submitting their final-year project reports. Please complete the checklist below to provide transparency regarding the use of Generative AI tools during the project work. Submit this form alongside your final report.

Student Details:

- Name: Rayaan Rafique
- Student ID: M00976008
- Project Title: ARiS – AI Driven Accessibility Risk Mapping and Intelligence System
- Supervisor: Mr. Roshan Renji

Generative AI Usage Declaration

Please indicate how generative AI tools were used during your project by ticking the relevant boxes below:

- ☐ No Generative AI tools were used during the completion of this project.
- ☐ AI tools were used for brainstorming and idea generation.
- ☐ AI tools were used for text drafting and report writing assistance.
- ☒ AI tools were used for code generation, debugging, or suggestions.
- ☒ AI tools were used for data analysis or visualisation support.
- ☐ AI tools were used for design simulation, modelling, or testing.
- ☐ Other (Please specify): _____

Acknowledgment

I hereby confirm that:

- All AI-generated content used in this report was critically evaluated, edited, and verified for accuracy and appropriateness.
- All intellectual contributions remain my own, and the final report reflects my independent work and understanding.
- Proper acknowledgment has been given where applicable.

Student Signature: Rayaan Rafique Date: 22/04/2026